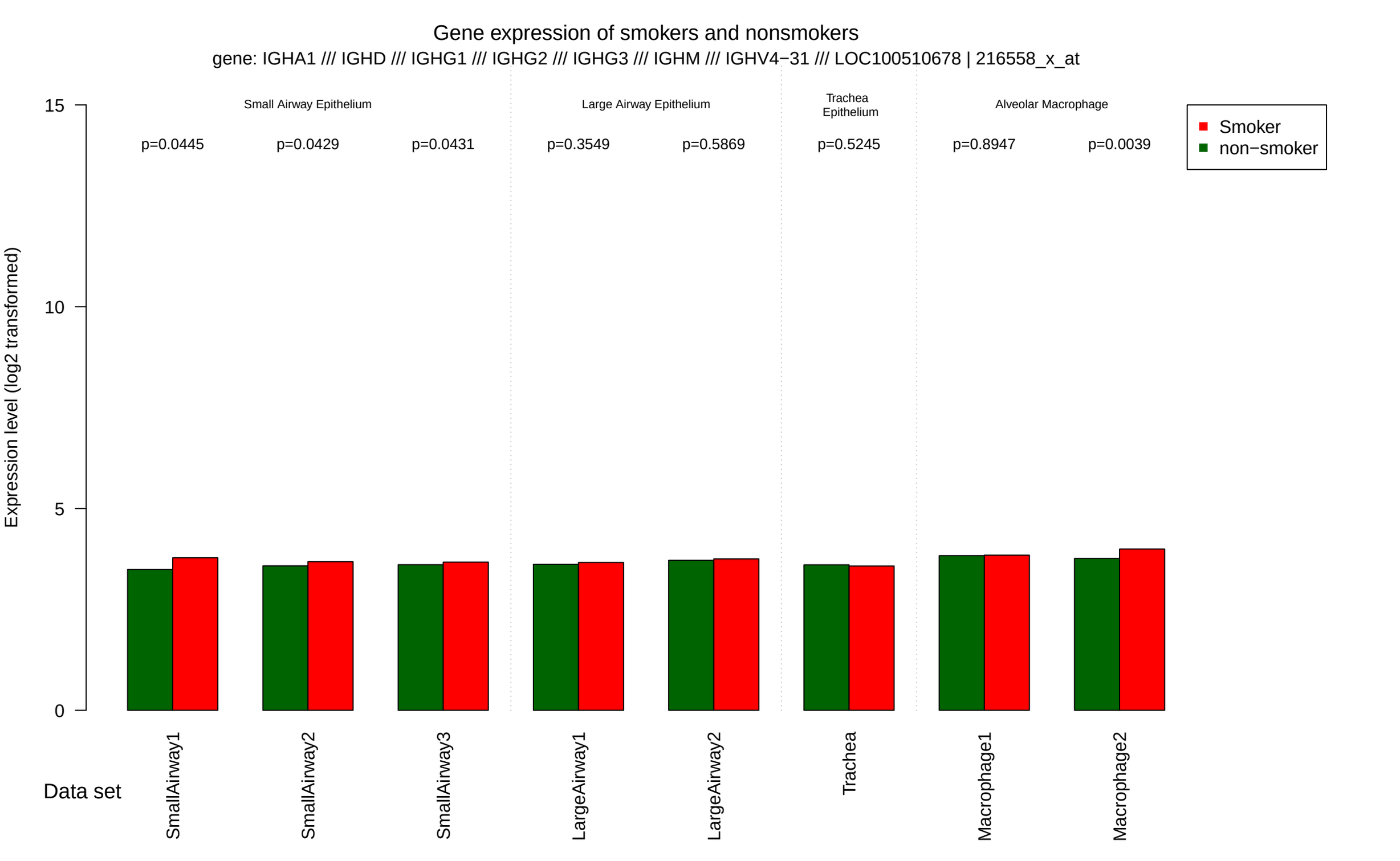
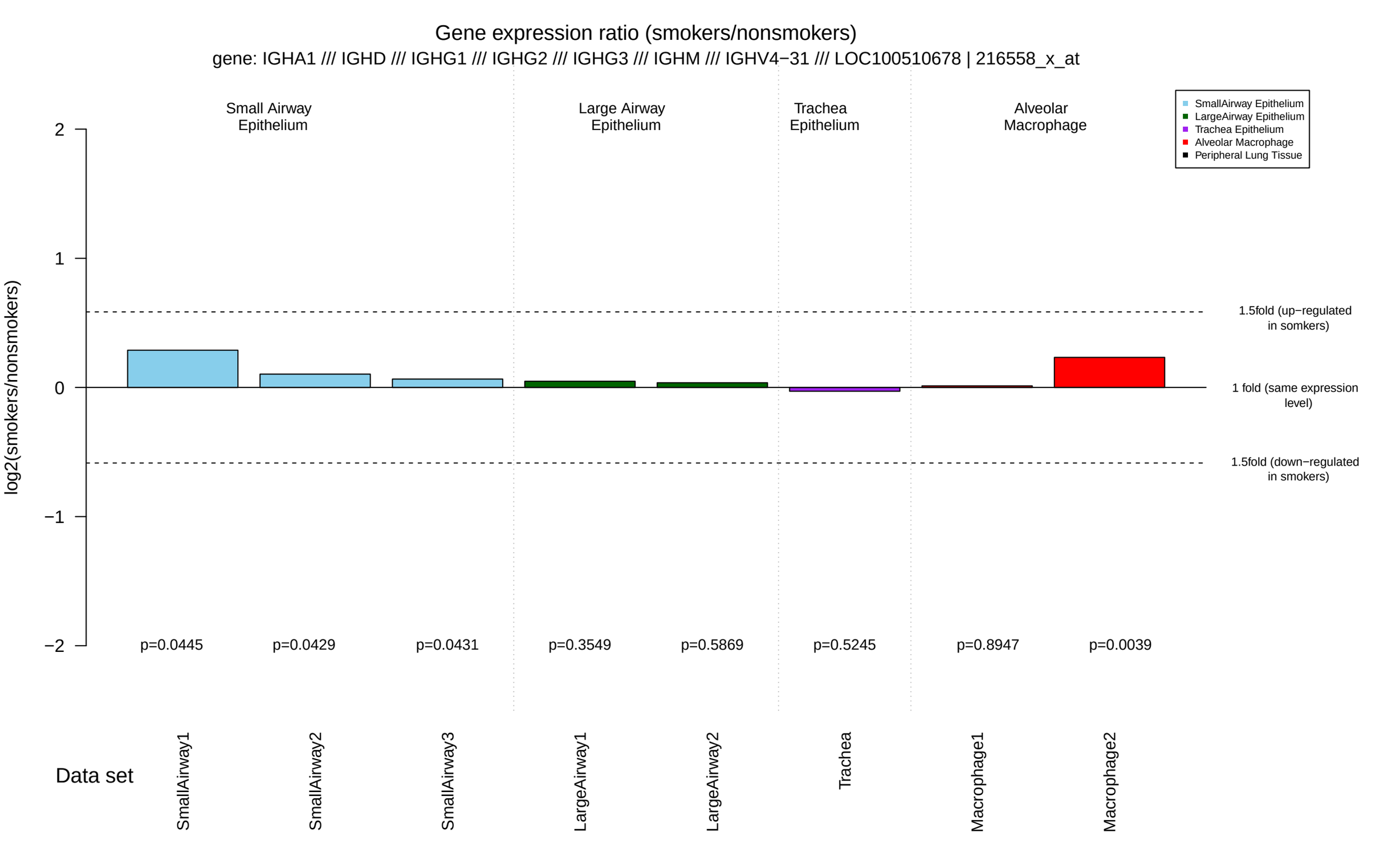
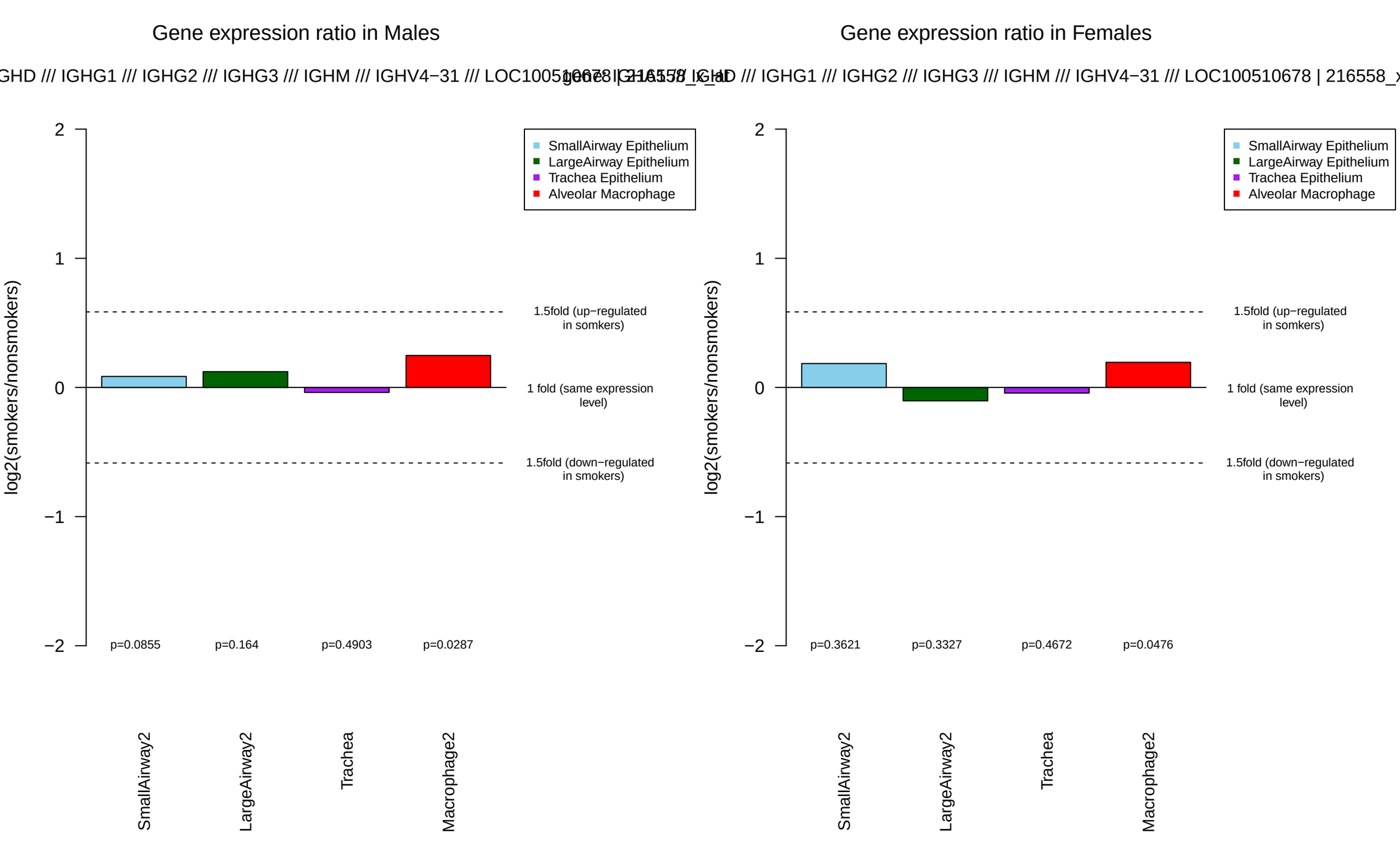
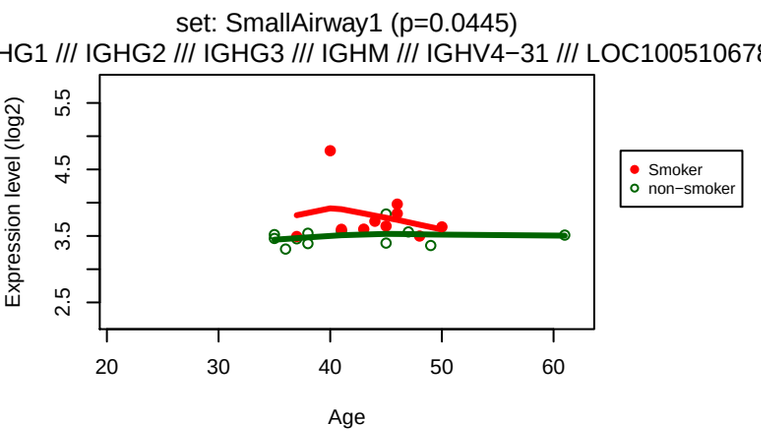
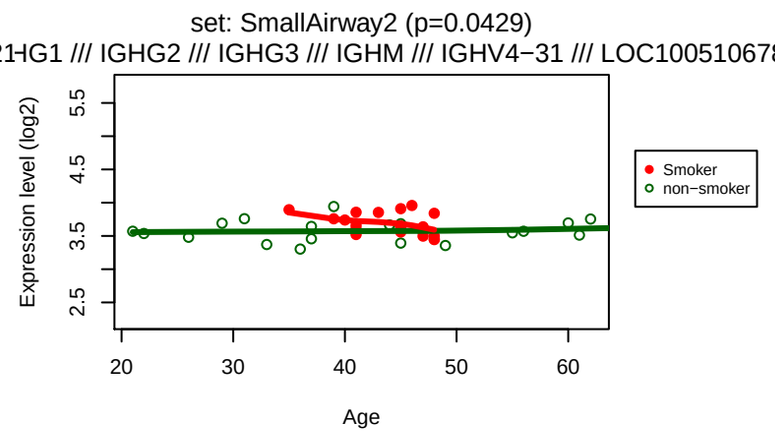
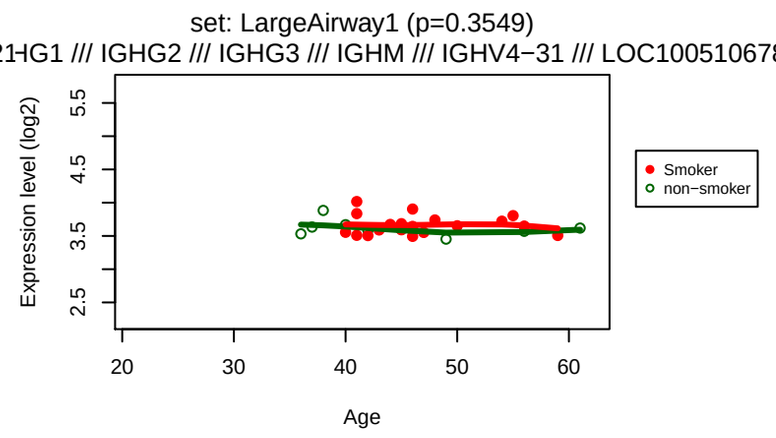

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHD /// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 216558_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0445
p=0.0429
p=0.0431
p=0.3549
p=0.5869
p=0.5245
p=0.8947
p=0.0039
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

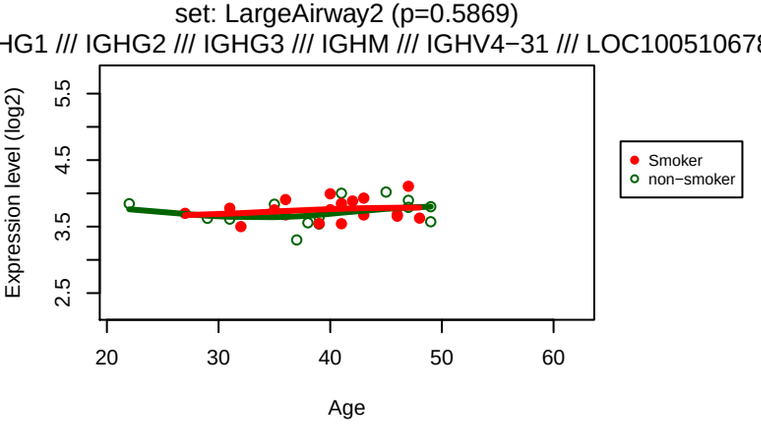

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHD /// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 216558_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0445
p=0.0429
p=0.0431
p=0.3549
p=0.5869
p=0.5245
p=0.8947
p=0.0039
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

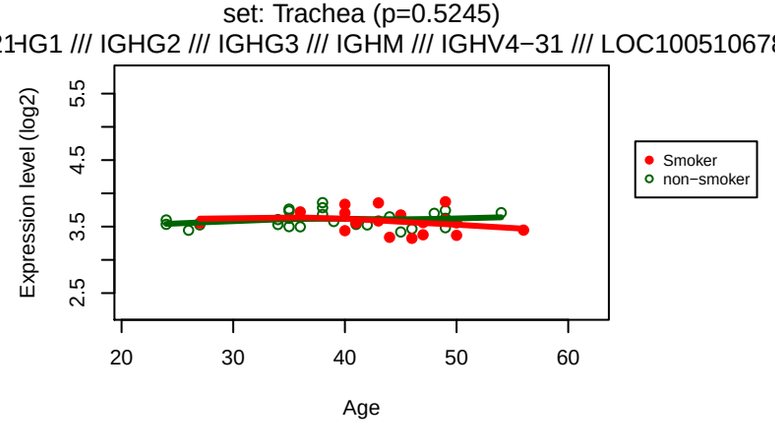

Gene expression ratio in Males
IGHD /// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 216558_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0855
p=0.164
p=0.4903
p=0.0287
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
IGHD /// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 216558_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3621
p=0.3327
p=0.4672
p=0.0476
SmallAirway2
LargeAirway2
Trachea
Macrophage2

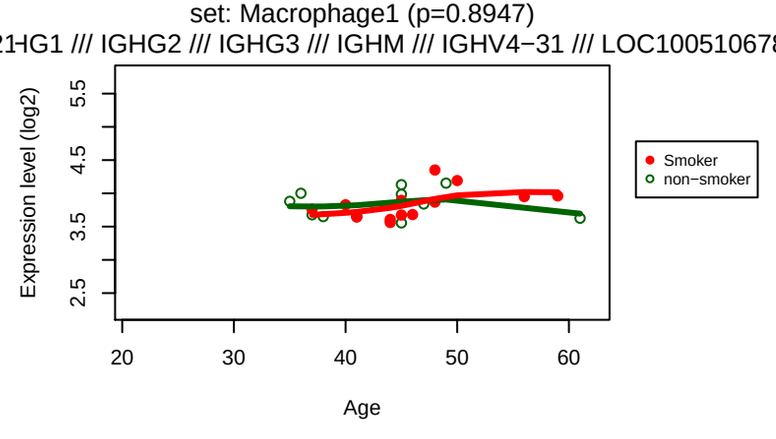

set: SmallAirway1 (p=0.0445)
HG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 21
Expression level (log2)
Age
Smoker
non−smoker

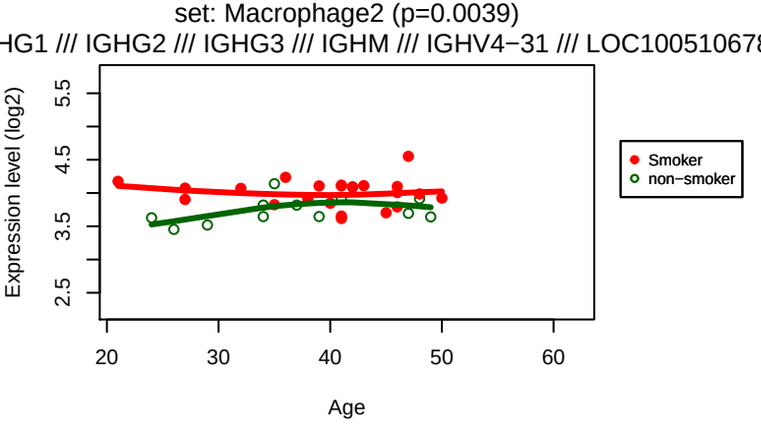

set: SmallAirway2 (p=0.0429)
HG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 21
Expression level (log2)
Age
Smoker
non−smoker

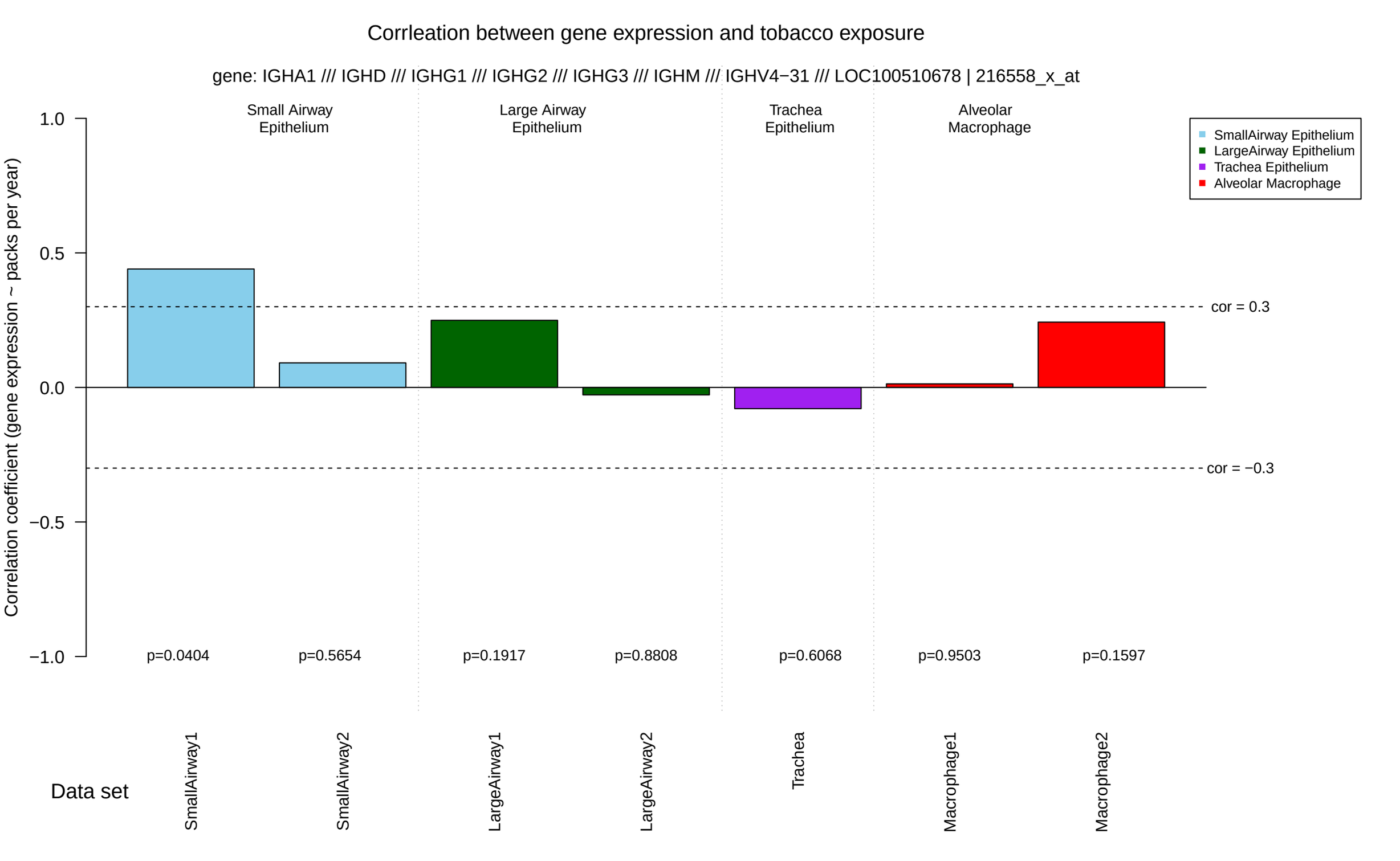

set: LargeAirway1 (p=0.3549)
HG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 21
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5869)
HG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 21
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5245)
HG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 21
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8947)
HG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 21
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0039)
HG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 21
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHD /// IGHG1 /// IGHG2 /// IGHG3 /// IGHM /// IGHV4−31 /// LOC100510678 | 216558_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0404
p=0.5654
p=0.1917
p=0.8808
p=0.6068
p=0.9503
p=0.1597
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2